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Linear resonant electrostatic induction motor using electrical resonance with piezoelectric transducers

Ryosuke Saito, Takuya Hosobata *, Akio Yamamoto, Toshiro Higuchi

Department of Precision Engineering, School of Engineering, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-8656, Japan

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ABSTRACT

This paper proposes an electrostatic linear induction motor resonating electrically with piezoelectric transducers. The motor is based on an electrostatic film motor composed of two thin plastic films containing fine pitch three-phase electrodes, which face each other across a short gap. The motor requires three-phase high voltages on the electrodes in both films to obtain thrust force. Our proposal in this paper is to utilize piezoelectric transducers to realize the high voltage on one of the films: the slider film, by using the transducers as inductive elements that resonate with the electrostatic motor. We describe specific design method of the piezoelectric transducers for this application, and analyze the influences of their properties on the performance of the motor. A working prototype was constructed with piezoelectric length-extensional bars as the transducers. The prototype demonstrated a maximum thrust force of 0.24 N with 500 $V_{0\text{-}p}$ three-phase excitation. When driven at high voltages, we observed a step-like motion caused by a non-linear behavior of the piezoelectrics, i.e., the jump resonance. The observation indicates a problem in optimal design of the piezoelectric transducers for this specific application.

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1. Introduction

Electrostatic motors require fine electrode structure for good performance. One idea of utilizing this motor is to downsize the overall structure to microscopic level, which has been applied to actuators for MEMS (MicroElectroMechanical Systems). Another idea is to integrate hundreds and thousands of fine-pitched electrodes to form a macro-scale motor. This paper concerns the latter type of electrostatic motor; we introduce piezoelectric transducers to resonate with a macro-scale electrostatic linear motor, in order to produce high voltages that are necessary for large thrust force. The piezoelectric transducers enable the motor to operate without direct feeding of power to its slider; the absence of power cables allows the slider to travel long distance without disturbance. Further, because both the piezoelectrics and the electrostatic motor are non-magnetic, the whole actuator system is non-magnetic. Thus, the motor provides long-stroke linear drive that can be used in MRI (Magnetic Resonance Imaging) devices, in which the use of magnetic motors are prohibited.

The motor we propose originates in an electrostatic linear motor reported in Ref. [1], which is illustrated in Fig. 1(a). The motor is composed of a pair of plastic films with fine pitch electrodes, facing each other across a short gap. The motor is driven synchronously by applying three-phase ac high voltages (in order of kilo volts) to the electrodes in both films. The voltages excite a strong electric field in the gap and realize high power-to-weight ratio; e.g., the motor in Ref. [1] had output performance of 230 W/kg. One remarkable feature of the motor is in its non-magnetism; an open-loop position control of this motor in a strong magnetic field of an MRI scanner was demonstrated in Ref. [2]. One disadvantage of this motor is that both of the films require cables for direct feeding of power. Such cables disturb dynamic motion of the slider: the traveling film; thus, the motor is unsuitable for some systems requiring cable-free sliders, such as multi-DOF actuators [3].

The rotary motor in Ref. [4] showed one solution which can free the slider from cables. In the motor, the voltages are supplied only to the stator electrodes and the rotor electrodes are connected to coils, which resonate with the capacitance of the electrodes for voltage boosting. The motor is an asynchronous motor and thus referred to as a resonant electrostatic induction motor.

The name "electrostatic induction motor" was originally given to motors with their rotors made from dielectrics coated with slightly conductive materials [5–7]. In these motors, the rotor gains torque when put in a rotating electric field formed by exciting the electrodes in the stator. Characteristics of these motors are dependent on the slip: relative speed of the field to the rotor. The resonant electrostatic induction motor is also slip dependent,

* Corresponding author. Tel.: +81 3 5841 6466; fax: +81 3 5841 3018.
E-mail addresses: ryosuke_saito@aml.t.u-tokyo.ac.jp (R. Saito), hosobata@aml.t.u-tokyo.ac.jp (T. Hosobata), akio@aml.t.u-tokyo.ac.jp (A. Yamamoto), higuchi@aml.t.u-tokyo.ac.jp (T. Higuchi).

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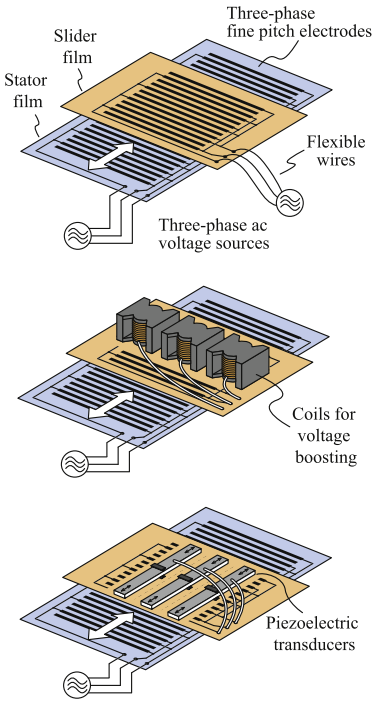


Fig. 1. Conceptual illustration of three electrostatic film motors: (a) a synchronous electrostatic motor with dual excitation, (b) a resonant electrostatic induction motor with mounted coils and (c) the same motor with coils replaced by piezoelectric transducers.

but differs greatly from other electrostatic induction motors in the rotor structure; it has electrodes embedded in them. The structure allows connection of electrical elements, such as coils for modification of characteristics.

As was demonstrated with a linear resonant electrostatic induction motor in Ref. [8], when the coils are mounted on the slider as in Fig. 1(b), the slider could be released from electrical connections to external equipments. However, by having the coils, the motor has lost the great advantage of electrostatic motors: the non-magnetism.

Fig. 1(c) illustrates a concept we proposed in our previous work to realize a non-magnetic resonant electrostatic induction motor [9]. In this motor, the coils are replaced with piezoelectric transducers. Such replacement is possible because piezoelectric transducers behave inductively near resonance. The inductive behavior comes from the inertia of the oscillating transducers; hence they provide electrical inductance without using magnetic materials nor producing magnetic field. By having this non-magnetic inductance to resonate with the capacitance of the electrodes, the motor can operate with large force while maintaining its non-magnetic property. Further, as shown in the figure, mounting the transducers on the slider realizes linear drive with no power cables connected to the slider. These two characteristics make the motor suitable for devices used in MRI scanners, especially for those requiring long-stroke linear drive. In this paper, we report an analysis which relates the properties of the piezoelectric transducers to the motor's performance. Our previous work in Ref. [9] explained the design concept of the motor based on admittance characteristics of the transducers and proved the feasibility of the idea with a working prototype. However, details on the design method considering the performance of the motor were not provided. Thus, in this work we provide detailed analysis to establish a design method for this motor.

The outline of this paper is as follows. First, in Section 2, we explain the principle of the resonant electrostatic induction motor and construct a model including piezoelectric transducers. Then, Section 3 describes the prototype of the motor, with emphasis on the design of the piezoelectric transducers. Section 4 provides experimental results. In these experiments, to simplify experimental conditions, the transducers are placed beside the motor instead of being mounted on the slider. Measured results include the slider voltage, the thrust force and the slider motion. In the experiments, the prototype showed step-like motion in some conditions, which cannot be explained under linear assumption of the piezoelectrics. The cause of such motion is discussed in Section 5. Finally, a conclusion follows in Section 6.

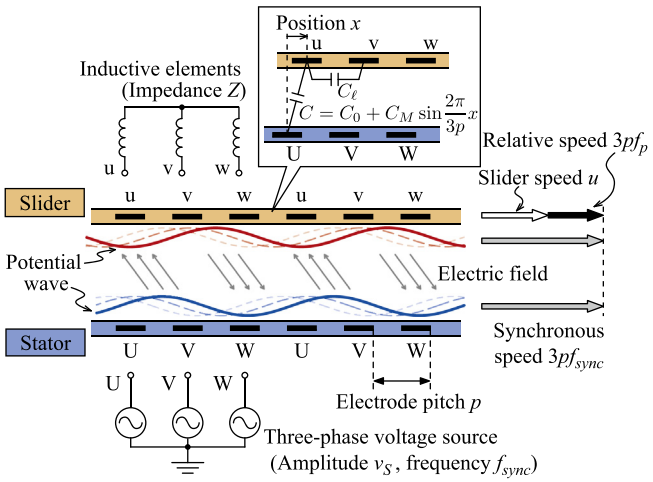


Fig. 2. Schematic showing the principle of a resonant electrostatic induction motor.

2. Modelling and analysis

This section explains a lumped-parameter model of the resonant electrostatic induction motor with piezoelectric transducers. We keep the explanations on the operating principle and the basic modeling brief and concentrate on the introduction of piezoelectrics; details on the basics are found in Ref. [4].

2.1. Principle of resonant electrostatic induction motor

Fig. 2 shows the principle of the resonant electrostatic induction motor. The motor is driven by a three-phase ac voltage applied to the three-phase electrodes in the stator film. The voltages excite an electric field traveling at a synchronous speed determined by the electrode pitch p and the excitation frequency f_{sync}. As the charges induced by the field travel on the slider, currents flow into the slider electrodes through the inductive elements. These currents induce voltages across the inductive elements, which form a potential wave on the slider. When this wave is displaced from the potential wave on the stator, the electric field exerts tangential forces to the charges on the slider and produces a thrust force.

The speed of the traveling field relative to the slider's actual speed has great influence on the motor's thrust force. The relative speed is represented by a frequency called the slip frequency f_p, which is defined as

$$f_p = f_{sync} - \frac{u}{3p}, \tag{1}$$

where u is the slider's speed. All voltages and currents in the slider side alternate with this frequency and the impedances of the inductive elements change accordingly.

Supposing that inductive elements with impedance Z are connected in Wye configuration, the voltage on one of the electrodes

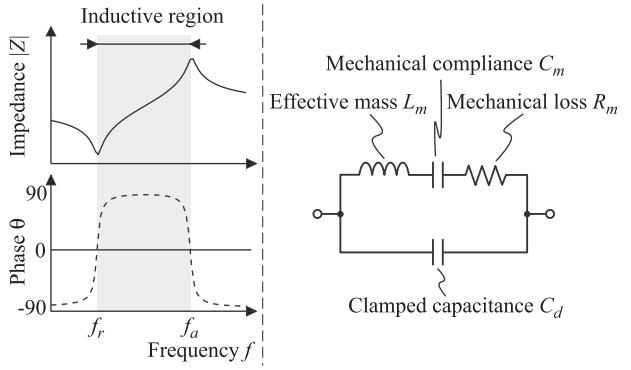


Fig. 3. Butterworth Van Dyke model of a typical piezoelectric transducer [10]. The transducer acts as an inductor only between the resonance and anti-resonance frequencies.

in the slider ($\widehat{V}_L$) can be expressed as a function of the slip angular frequency $\omega_p = 2\pi f_p$, which is given as

$$\widehat{V}_L = \frac{3}{2} C_M \frac{j\omega_p Z}{1 + j\omega_p C_p Z} v_S. \tag{2}$$

Voltages on the other two electrodes are calculated from this voltage, by shifting the phase by ±120°. Symbols in the equation represent amplitude of excitation voltage (v_S), amplitude of position dependent capacitance (C_M) and a sum of constant capacitances of the slider electrodes ($C_p = 3(C_\ell + C_0)$). From $\widehat{V}_L$, the thrust force of the motor is calculated as

$$F = \frac{3\pi}{2p} C_M v_S v_L \sin\phi, \tag{3}$$

where $v_L = |\widehat{V}_L|$ and $\phi = \angle \widehat{V}_L$. The amplitude v_L corresponds to the amplitude of the potential wave on the slider, and the phase ϕ corresponds to the spatial phase difference between the two potential waves. This expression indicates that the thrust force becomes larger with higher slider voltage and the phase closer to $\pi/2$.

2.2. Consideration of frequency dependence of piezoelectrics

Piezoelectric transducers are often utilized as inductors in oscillation circuits. This work also focuses on using the inductive behavior of the transducers for resonating with the electrostatic motor. Fig. 3 shows an example of an impedance curve of a piezoelectric transducer, around its resonance state. The transducer can act as an inductive element between its resonance and anti-resonance frequencies where the phase of its impedance is positive. Also shown in Fig. 3 is an equivalent circuit model of the piezoelectric transducer, widely known as the Butterworth Van Dyke (BVD) model [10], which represents the electrical behavior of the transducer near resonance.

In the BVD model, the top branch is the mechanical equivalent circuit, where L_m, C_m and R_m represent an equivalent mass, compliance and a mechanical damping loss, respectively. By introducing a mechanical quality factor Q_m, the mechanical loss R_m can be expressed as

$$R_m = \frac{1}{Q_m}\sqrt{\frac{L_m}{C_m}}. \tag{4}$$

The capacitance element C_d in the lower branch indicates an electrical capacitance referred to as a clamped capacitance. By using this model, the electrical impedance of the transducer can be represented by

$$Z = \left(j\omega C_d + \frac{1}{R_m + j\omega L_m + \frac{1}{j\omega C_m}} \right)^{-1}. \tag{5}$$

With Eqs. (2) and (5), voltages of the slider electrodes are expressed in terms of parameters of the piezoelectric transducers. First, we define a combined capacitance of the slider electrodes and the transducer as

$$C_e = \left(\frac{1}{C_m} + \frac{1}{C_d + C_p} \right)^{-1}. \tag{6}$$

Using Eq. (6), a resonance frequency between the combined capacitance and the inductance of the transducer is defined as

$$\omega_n = \sqrt{\frac{1}{L_m C_e}}. \tag{7}$$

Using these expressions, the voltages of the slider electrodes are obtained as

$$\widehat{V}_L = \frac{3C_M}{2(C_d + C_p)} \frac{\sqrt{\frac{L_m}{C_e}}\left(\frac{C_e}{C_m} - \frac{\omega_p^2}{\omega_n^2}\right) + jR_m\frac{\omega_p}{\omega_n}}{\sqrt{\frac{L_m}{C_e}}\left(1 - \frac{\omega_p^2}{\omega_n^2}\right) + jR_m\frac{\omega_p}{\omega_n}} v_S. \tag{8}$$

At the resonance condition $\omega_p = \omega_n$, the amplitude and the phase of the slider voltages become

$$v_L = \frac{3C_M}{2(C_d + C_p)} \sqrt{1 + \frac{L_m C_m}{R_m^2 (C_d + C_p)(C_m + C_d + C_p)}} v_S, \tag{9}$$

and

$$\phi = \tan^{-1} \frac{1}{R_m} \sqrt{\frac{L_m C_m}{(C_d + C_p)(C_m + C_d + C_p)}}. \tag{10}$$

As shown in Eq. (3), maximum v_L and $\phi = \pi/2$ maximize the thrust force. Considering Eqs. (9) and (10), low R_m and C_d in addition to high L_m and C_m are preferred to approach such condition. Among these parameters, R_m and C_d have significant influences on v_L. Details on how to determine these parameters are described in Section 3.

2.3. Inclusion of dielectric loss in slider

Analysis in the last section indicates the importance of low mechanical loss in the piezoelectric transducers. Similarly, dielectric loss in the slider electrodes may also degrade the quality of resonance, and thus need to be included in the model. Generally, dielectric loss of a capacitance C is represented as a conductance

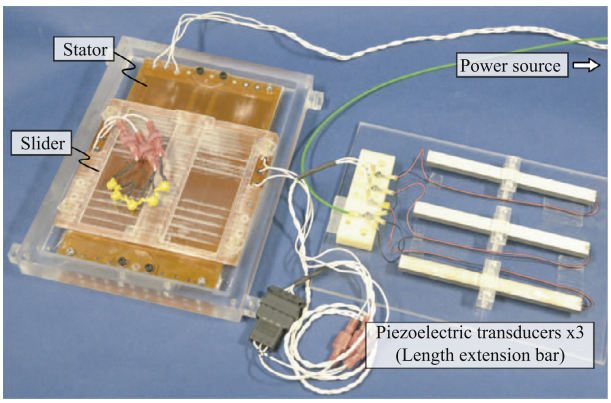


Fig. 4. A linear resonant electrostatic induction motor with three piezoelectric transducers connected in a Wye configuration.

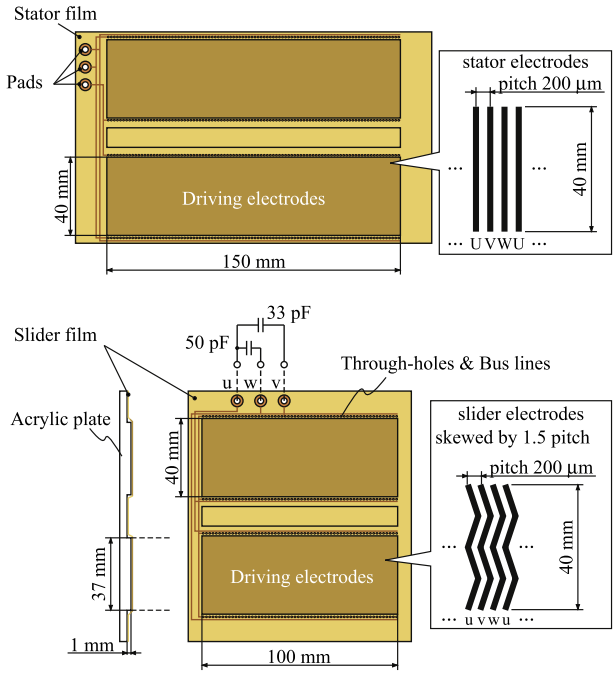


Fig. 5. Dimensions of the stator and the slider film.

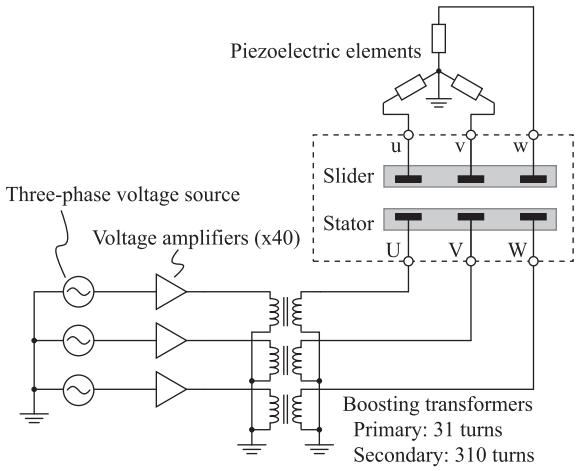


Fig. 6. Schematic of the driving circuit. The three-phase signal was generated with a function generator (7075, HIOKI), which was first amplified by 40 times, using three voltage amplifiers (4020, NF). The amplified voltages were further amplified by ten times using three transformers, wound on a ferrite core (PQ5050, TDK).

G and a dielectric loss tangent of the material $\tan\delta$ relates them as $G = \omega C \tan\delta$. By introducing a dielectric loss G_p corresponding to the capacitance C_p in Eq. (2), the slider voltage is expressed as

$$\hat{V}_L = \frac{3}{2} C_M \frac{j\omega_p Z}{1 + (G_p + j\omega_p C_p)Z} v_S. \tag{11}$$

By substituting Eq. (5) into Eq. (11), the slider voltage is calculated as

$$\hat{V}_L = \frac{K\left\{\sqrt{\frac{L_m}{C_e}}\left(\frac{C_e}{C_m} - \frac{\omega_p^2}{\omega_n^2}\right) + jR_m\frac{\omega_p}{\omega_n}\right\} v_S}{\sqrt{\frac{L_m}{C_e}}\left(1 - \frac{\omega_p^2}{\omega_n^2}\right) + \frac{R_m G_p}{\omega_n C_K} + j\frac{\omega_p}{\omega_n}\left\{R_m + \sqrt{\frac{L_m}{C_e}}\frac{G_p}{\omega_n C_K}\left(1 - \frac{C_e\omega_n^2}{C_m\omega_p^2}\right)\right\}}, \tag{12}$$

where $C_K = C_d + C_p$ and $K = 3C_M/2C_K$.

3. Design of the prototype motor

This section explains the details of each component in the prototype motor shown in Fig. 4. The components are the two films, the driver electronics, and the three piezoelectric transducers. In this work, we focus on designing the piezoelectric transducers to match the other two existing components.

3.1. Stator film and slider film

The films used in the prototype are shown in Fig. 5. The films are flexible printed circuits, with polyimide as the material for both the substrate and the cover. The stator film contains two areas of three-phase electrodes for generating thrust force, which are straight copper lines equally spaced with a pitch of 200 μm. The slider film also has two areas of three-phase electrodes with the same pitch, but its electrodes are skewed by 1.5 times the pitch; the skewing reduces the ripple of the thrust force [11]. Thicknesses of the films are 200 μm.

The two films are faced across a small gap and operated by applying three-phase ac high voltages such as 1 $kV_{0\text{-}p}$. Such high voltages cause corona discharge when the motor is operated in the air. To prevent the discharge, the gap between films is filled with dielectric liquid (Fluorinert FC-77, 3M). Further, to maintain a constant gap and to reduce friction, glass beads with a diameter of 20 μm are spread between the films.

The slider film is fixed to a convex shaped plastic board and two external capacitors (33 pF and 50 pF) are connected to its electrodes for correction of a capacitance unbalance. With the unbalance corrected, capacitance parameters were identified as $C_p = 2.12$ nF and $C_M = 90$ pF. The dielectric loss tangent $\tan\delta_p$, corresponding to C_p was also identified as 0.026.

3.2. Driver electronics

Fig. 6 shows the electrical connection of the components. To obtain three-phase ac high voltage, the output of a signal generator (7075, HIOKI) is first amplified by voltage amplifiers (4020, NF) and then by boosting transformers. The purpose of utilizing transformers is to reduce the reactive current. The secondary inductances of the transformers and the capacitances of the stator electrodes were approximately 67 mH and 3.2 nF, respectively. With secondary voltage of transformers boosted up to 1 $kV_{0\text{-}p}$, the amplifier could drive the load at frequencies from 8.0 kHz to 14.5 kHz, within its average current limit of 1.0 A.

3.3. Design of piezoelectric transducers

The first step in designing the piezoelectric transducers was to choose the piezoelectric material. As explained in Section 2, the piezoelectric transducers should have high mechanical quality factor Q_m and low clamped capacitance C_d to obtain high voltage on the slider electrodes. Among piezoceramic materials, hard

Table 1
Properties of hard PZT material C-213 manufactured by Fuji Ceramics Corp. Parameters used in this work are extracted from their catalog [18].

Coupling factor	k_{31}	0.34
Frequency constant	N_{31} [m Hz]	1620
Dielectric constant	$\varepsilon_{33}^T/\varepsilon_0$	1470
Mechanical Q	Q_m	2500
Piezoelectric charge constant	d_{31} [C/N]	-135×10^{-12}
Dissipation factor	$\tan\delta$ [%]	0.30
Young's modulus	$Y_{11}^E = 1/s_{11}^E$ [N/m^2]	8.2×10^{10}
Density	ρ [kg/m^3]	7.8×10^3

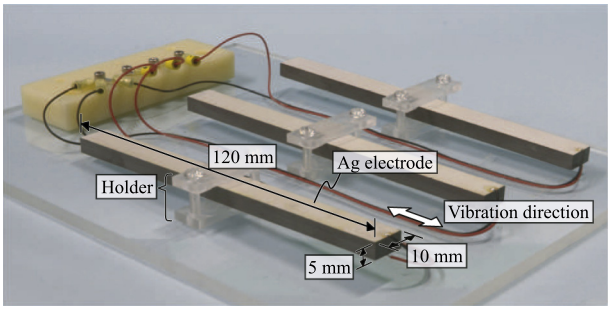


Fig. 7. Dimensions of the designed piezoelectric transducers. The transducers are operated in fundamental length extensional mode with transverse excitation, with holders clamping their vibration nodes.

ceramics are known to have higher quality factor and lower permittivity than soft ceramics. Thus, we chose hard PZT material C-213 (Fuji Ceramics Corp.), whose properties are listed in Table 1. Using this material, we designed the transducers as shown in Fig. 7. The transducers are rectangular bars, operated in the fundamental length extensional mode with transverse excitation. To maintain high mechanical quality factor, it is also important to prevent the leakage of vibration energy through supporting points. Thus, to minimize the leakage, we designed holders to clamp the transducers at their vibration nodes, which appear at the center of the length.

Dimensions of the transducers were designed in following steps. First, we determined the length of the transducers considering the driving frequency. The driving frequency must be within the drivable range of the amplifiers and the transducers must behave inductively near this frequency. Thus, the resonance frequency (f_r) and anti-resonance frequency (f_a) of the transducers must be considered first. For a length extensional piezoelectric transducer operated in a fundamental mode, their length ℓ is directly related to its resonance and anti-resonance frequencies, which are expressed as

$$f_r = \frac{1}{2\ell\sqrt{\rho s_{11}^E}}, \quad f_a = f_r\sqrt{1+\frac{8}{\pi^2}\frac{k_{31}^2}{1-k_{31}^2}}, \tag{13}$$

where ρ is a density, s_{11}^E is an elastic compliance constant and k_{31} is a coupling constant. From the coupling constant of the material, we can calculate $f_a = 1.05 f_r$; the anti-resonance frequency is slightly higher than the resonance frequency. We designed the transducers with their lengths 120 mm to have the frequencies at $f_r = 13.5$ kHz and $f_a = 14.2$ kHz, which is within the drivable range of the amplifiers.

In the next step, we determined the aspect ratio α, which is defined as the width (w) of the transducer divided by its thickness (t), because the ratio is related to the amplitude of slider voltage at resonance. Parameters of the BVD model of the transverse length extensional transducer can be estimated from material properties and dimensions [12]. The equivalent mass L_m, compliance C_m and the clamped capacitance C_d are respectively expressed in terms of the aspect ratio α as

$$L_m = \frac{\rho}{8}\frac{{s_{11}^E}^2}{d_{31}^2}\frac{\ell}{\alpha}, \quad C_m = \frac{8}{\pi^2}k_{31}^2\varepsilon_{33}^T\ell\alpha, \quad C_d = (1-k_{31}^2)\varepsilon_{33}^T\ell\alpha, \tag{14}$$

with symbols d_{31}, k_{31} and ε_{33}^T representing piezoelectric constant, coupling coefficient and permittivity. In addition to these parameters, the mechanical loss R_m must be estimated to calculate the slider voltage. However, precise estimation is difficult because R_m is influenced by many factors such as acoustic radiation to the atmosphere and energy leak from supporting parts. Thus, we simply calculated the value from the mechanical quality factor by substituting Eq. (14) into Eq. (4) as

$$R_m = \frac{1}{Q_m}\frac{\pi\sqrt{\rho s_{11}^E}}{8k_{31}^2\varepsilon_{33}^T}\frac{1}{\alpha}. \tag{15}$$

Based on these parameters, we calculated the stator-to-slider voltage-boosting ratio (v_L/v_S) by Eq. (12) with various aspect ratios (α), and the result is shown in Fig. 8. The amplitude of the slider voltage becomes larger with lower α and maximized around $\alpha = 1.4$, which is the optimal value for this particular design. We chose a slightly higher value $\alpha = 2$ in the actual design.

Finally, the absolute value of the thickness t was determined considering the electric field within the piezoelectric transducers; thinner the transducer is, stronger the electric field. The manufacturer of the material recommends maximum electric field for continuous use to be within 400 V/mm. Thus, we chose 5 mm as the thickness (hence 10 mm as the width), so that the transducers withstand high voltages up to 2 kV. Note that this design step needs reconsideration to avoid non-linear region of the

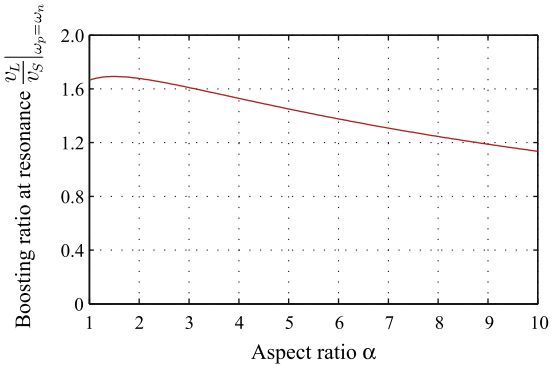


Fig. 8. Calculated voltage booting ratio of the slider voltage v_L to the excitation voltage v_S at resonance condition. The aspect ratio α is a ratio of the width to the thickness of the transverse length extensional transducer.

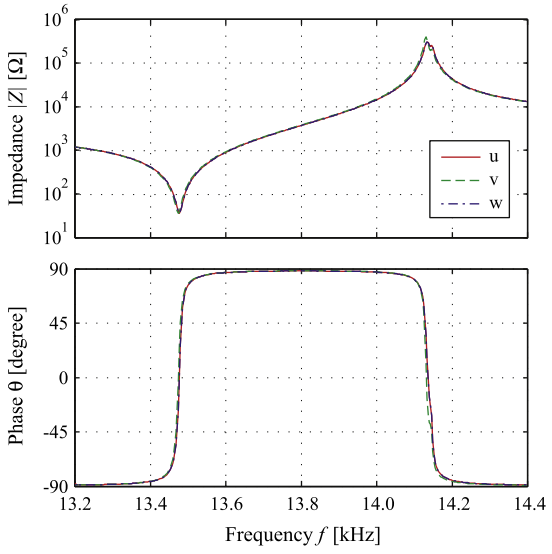


Fig. 9. Measured impedance of the designed piezoelectric transducers. The result shows that the resonance and anti-resonance frequencies are approximately at 13.5 kHz and 14.1 kHz, respectively.

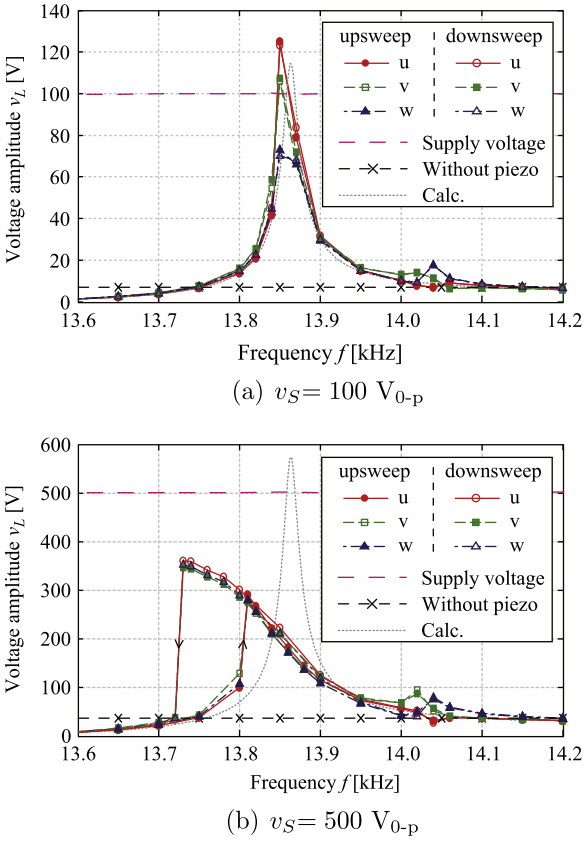


Fig. 10. Amplitudes of the voltages of the slider electrodes with piezoelectric transducers: (a) with 100 $V_{0\text{-}p}$ excitation and (b) with 500 $V_{0\text{-}p}$ excitation on the stator. Theoretical curves are calculated from Eq. (12).

transducers at high voltage excitations; details on this problem are provided in later sections.

Fig. 9 shows impedance curves of the designed transducers near the fundamental mode, which were measured by an impedance analyzer (4194, Hewlett Packard). The resonance and anti-resonance frequencies observed from this result are approximately at 13.5 kHz and 14.1 kHz, which are in good agreement with the estimated values. Between these frequencies, phase angle of the impedance is near 90°, which indicates that the transducers are operated with low loss.

From the measured impedance, the parameters in the BVD model are identified as $L_m = 477$ mH, $C_m = 292$ pF, $R_m = 32.9\ \Omega$ and $C_d = 2.87$ nF. Consequently, from Eq. (7), the resonance between the electrostatic motor and the piezoelectric transducers is expected to occur at 13.86 kHz.

4. Experimental results

4.1. Measurement of slider voltages

To confirm that the voltages of the slider electrodes are boosted by resonance with the piezoelectric transducers, we measured the amplitudes of the voltages with various excitation frequencies. The frequency was changed both in sweeping up and down. For comparison, the voltages without the transducers were also measured. The voltages were measured with the slider keeping its position; in this condition, the slip frequency f_p is equal to the excitation frequency f_{sync}. Additionally, we put a weight of 1 kg on top of the slider to ensure that the gap between the films is kept constant throughout the voltage measurement.

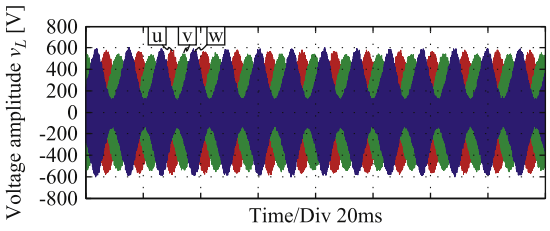


Fig. 11. Beat waves of the slider voltages at an excitation of 500 $V_{0\text{-}p}$/13.75 kHz in a downward sweep. The Nyquist frequency of the oscilloscope was configured at 25 kHz.

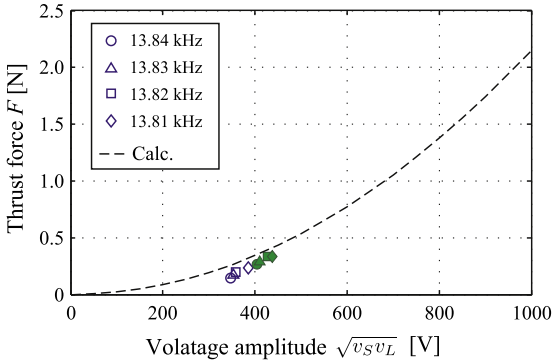


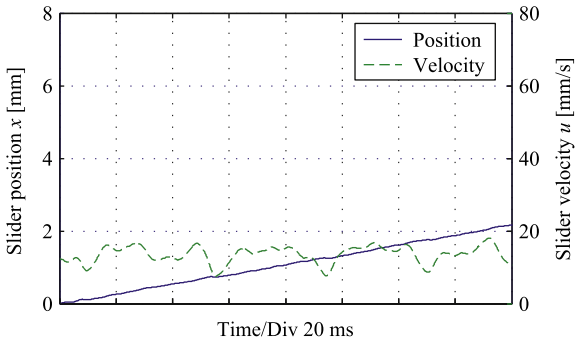
Fig. 12. Comparison of measured thrust force with a theoretical value.

Fig. 10 shows the measured voltage amplitudes on each slider electrode with 100 $V_{0\text{-}p}$ and 500 $V_{0\text{-}p}$ excitations. Each graph includes calculation values using Eq. (12), based on the identified parameters. Without the piezoelectric transducers, the slider voltages were ten times smaller than the excitation voltage, which are insufficient to drive the motor. In the case with 100 $V_{0\text{-}p}$ excitation, connecting the transducers boosted the voltages at resonance by about 17 times. However, even with the boosting, the voltages were too low to drive the motor. With the excitation by 500 $V_{0\text{-}p}$ voltages, the slider voltages were boosted by about ten times, but the peaks of voltage amplitudes appeared at frequencies considerably lower than the theoretical expectation. Further, the voltage amplitudes followed different curves depending on the sweeping direction of the frequency. This is caused by the jump resonance [13], which is known to occur in the non-linear vibration systems with a soft spring, especially in high power applications of piezoelectrics such as ultrasonic motors [14] and piezoelectric transformers [15].

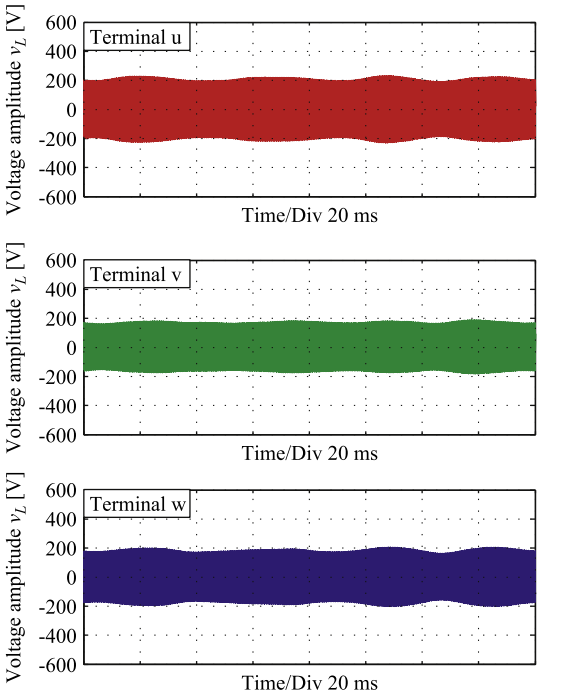
Another phenomenon observed despite our expectation was the beat of the slider voltages. This occurred with 500 $V_{0\text{-}p}$ excitation, during down-sweep from 13.81 kHz to 13.74 kHz. Fig. 11 shows an example of the beat waveforms measured at the excitation of 500 $V_{0\text{-}p}$/13.75 kHz. In the example, the envelopes of the three beat waves appear as a three-phase ac waveform. The beat frequency is approximately 88 Hz and this frequency was almost the same regardless of the excitation frequency. We presume that the beat is also coming from the non-linearity of the piezoelectric transducers under the strong electric field, because the beat never occurred with lower excitations voltages. With 500 $V_{0\text{-}p}$ excitation, the beat was present just for a short time after the start of voltage application, whereas with higher voltages it stayed to exist. For this reason, the excitation voltages in all experiments were limited to 500 $V_{0\text{-}p}$.

4.2. Measurement of thrust force

The thrust force of the prototype motor was measured by recording the amount of weight it could hold. In the measurement,

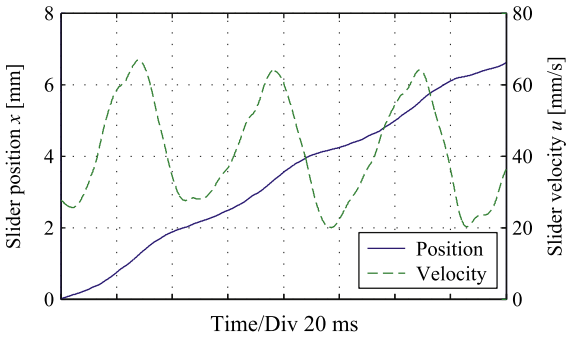


(a) Position and velocity of the slider

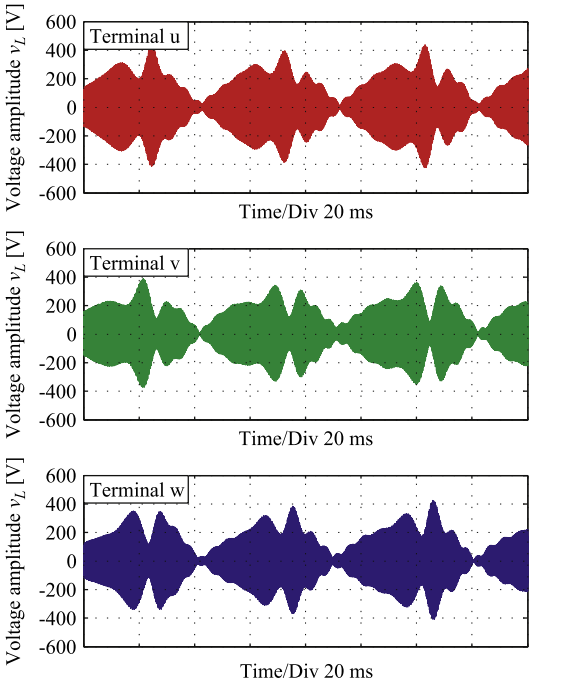


(b) Voltage waveforms of the slider electrodes

Fig. 13. Motion trajectory of the slider and voltage waveforms of the slider electrodes at excitation of 500 $V_{0\text{-}p}$/13.80 kHz. The position and velocity data were processed by zero-phase digital filtering using second-order Butterworth LPFs with cutoff frequencies of 200 Hz and 70 Hz, respectively.



(a) Position and velocity of the slider



(b) Voltage waveforms of the slider electrodes

Fig. 14. Step-like motion trajectory of the slider and voltage waveforms of the slider electrodes at excitation of 500 $V_{0\text{-}p}$/13.86 kHz. The position and velocity data were processed by zero-phase digital filtering using second-order Butterworth LPFs with cutoff frequencies of 200 Hz and 70 Hz, respectively.

one end of a string with weights on the other end was tied to the slider. With an excitation voltage applied, the weights were hanged down through a pulley to impose a force on the slider acting against its thrust force. At this condition, we recorded the maximum weight the motor could hold without traveling backwards. Amplitude of excitation voltage was set at 500 $V_{0\text{-}p}$, and the excitation frequency was always swept downward when adjusting.

Fig. 12 shows the measured forces, together with a calculated curve. The calculation is based on Eq. (3), assuming ideal condition of $\phi = \pi/2$ (resonance condition). From the equation, we expect the force to be proportional to the product of the stator and slider voltages. Thus, we plotted the forces with geometric mean of these voltages corresponding to the horizontal axis.

The maximum thrust force recorded was 0.24 N at 500 $V_{0\text{-}p}$/13.81 kHz. Larger thrust force could not be obtained because the excitation voltage was limited by the beating of voltages. If this problem is solved in the future, we expect the thrust force to increase along the theoretical curve, because we see a good agreement between the measured and theoretical values.

4.3. Trajectory of slider motion

The position of the slider was measured as it traveled with no load, using a laser displacement sensor (LB-01, Keyence). Fig. 13(a) shows the position and the velocity of the slider driven at an excitation of 500 $V_{0\text{-}p}$/13.80 kHz. In Fig. 13(b) are the voltages of the slider electrodes measured at the same time. In this condition, the slider smoothly traveled at a constant velocity of 14 mm/s, with amplitudes of slider voltages almost kept constant at 200 $V_{0\text{-}p}$.

Fig. 14 shows the result with a slightly higher excitation frequency: 500 $V_{0\text{-}p}$/13.86 kHz. With this frequency, we observed an unexpected behavior. The average traveling velocity of the slider was 40 mm/s, faster than the previous result, but there was a repetitive change of velocity, resulting in a step-like motion. Time intervals and distances of this step-like motion was approximately 50 ms and 2.1 mm, respectively. More discussion on this phenomenon is provided in the next section.

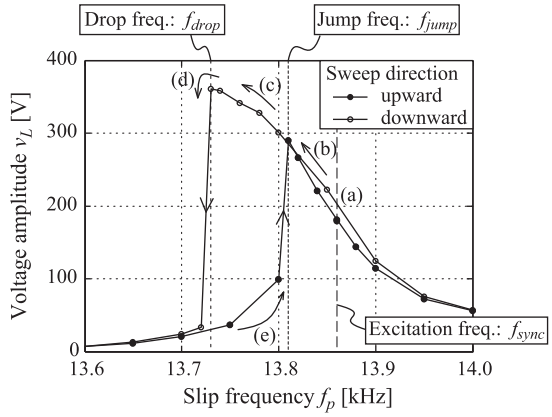


Fig. 15. A loop of slider voltages by jump resonance, causing step-like motion of the slider.

5. Discussion

This section discusses the problems found in the prototyping of the motor, which come from the non-linear behavior of the piezoelectrics that was not considered in the design. Especially, we focus on the cause of the step-like motion shown in Fig. 14(a); the strong non-linearity of the piezoelectric transducers is likely to be causing the step-like motion, by bringing beats to the slider voltages.

To avoid confusion, we make clear that there are two types of beating observed in the experiments of this work; the first appeared by applying high excitation voltage with the slider standing still (Fig. 11) and the second was observed during the traveling of the slider (Fig. 14(a)). The frequency of the former beat, observed at standstill, was approximately 88 Hz regardless of the excitation frequency, which is higher than the frequency of step-like motion (20 Hz). Thus, we draw a conclusion that the second type of beating is causing the step-like motion.

We can explain the cause of this beating by the jumping and dropping of the slider voltages observed in Fig. 10(b), and this characteristic provides an explanation to the step-like motion as shown in Fig. 15. Interpretation of the figure is as follows.

1. First, three-phase ac high voltage with a frequency f_{sync} is applied to the stator. At this moment, the slip frequency f_p is equal to the excitation frequency ($f_p = f_{sync}$) because the slider is standing still.
2. Then, the slider gains enough thrust force to overcome the friction between the films, and starts accelerating. As the velocity increases, the slip frequency decreases as shown in Eq. (1). At the same time, the slider voltages increase.
3. Increase of the slider voltages brings larger thrust force and further accelerates the slider and lowers the slip frequency.
4. When the slip frequency reaches the frequency f_{drop}, the slider voltages suddenly drop down, which results in a slowing down of the slider because the friction wins over the weakened thrust force.
5. As the slider decelerates, the slip frequency shifts to a higher frequency and the slider voltages rapidly jump up when the slip frequency reaches f_{jump}. Consequently, the slider regains a large thrust force due to the increase of the slider voltages.
6. The slider enters a loop, repeating from (c) to (e). Thus, the slider travels in a step-like trajectory.

As proved by the discussion above, the key to realize stable operation of this motor is in resolving the problems caused by the non-linearity of piezoelectric transducers. Three possible solutions are as follows. The first solution is to avoid the non-linear behavior by reducing the excitation field. This should be done by making the transducers thicker and not by lowering the voltage, as obtaining high voltage is the goal in design. The second solution is utilizing piezoelectric materials showing good linearity in high electric field, such as $LiNbO_3$ [16]. The third possible solution is to utilize piezoelectric transducers that behave as hard-spring oscillators; e.g., a $(Sr, Ca)_2NaNb_5O_{15}$ disk vibrating in radial direction [17]. Such oscillators show a resonance curve deformed towards the high frequency side; thus, the jumping and dropping frequencies where may be excluded from the motor's operating frequency.

6. Conclusion

This paper proposed an electrostatic induction motor that resonates with piezoelectric transducers acting as inductors. A basic principle of the motor and requirements for piezoelectric transducers were explained by using an equivalent circuit model. A working prototype was constructed and the motor's capability to operate with a thrust force up to 0.24 N was demonstrated with a 500 $V_{0\text{-}p}$ three-phase excitation. Although we succeeded in showing the effect of using piezoelectric transducers to raise the slider voltage and enhance the thrust force, we also found some unexpected problems. The problems are caused by non-linear behaviors of the piezoelectrics operated at high power. We expect that future studies solving these problems will lead to an establishment of complete design method of piezoelectric transducers for this specific application.

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